Plasmid preparation by alkaline lysis

Under alkaline conditions, high molecular weight DNA is selectively denatured while covalently closed circular plasmid DNA remains double-stranded. When the solution is rapidly neutralized, denatured chromosomal DNA forms an insoluble clot, while plasmid DNA remains in solution (Birnboim and Doly, 1979).

This method yields tens of micrograms of highly pure plasmid DNA from mini-preps (a few milliliters of culture), and up to milligrams from a liter-scale culture.

The plasmids are purified over a silica matrix (*Alternative A*) or by alcohol precipitation (*Alternative B*).

Risk assessment

▷ Wear gloves, safety glasses, lab coat

Reviewed: May 30, 2023

Procedures

›› *Alkaline lysis*

- ☐ Buffer P1 (Resuspension buffer), 250 mL ⟨*R*⟩
- ☐ Buffer P2 (Lysis buffer), 250 mL ⟨*R*⟩
- ☐ Buffer N3 (Neutralization buffer), 250 mL ⟨*R*⟩

(1.) Grow bacteria in 1–5 mL LB (Miller) overnight to stationary phase. Collect the cells by centrifugation.

> ***Hint:*** For culture tubes, spin at $3\,000 \times g$ for 5 min. For 2 mL centrifuge tubes, a brief spin at $11\,000 \times g$ is sufficient.

(2.) Resuspend the pellet in 150 µL Buffer P1.

> ***Hint:*** Buffers P1, P2, N3, PB, and PE refer to standard silica-column miniprep kit components (e.g., QIAGEN QIAprep). Equivalent buffers from other suppliers can be substituted. Homemade buffer recipes are available elsewhere.

(3.) Add 250 µL Buffer P2. Mix by gently inverting the tube 6–8 times. Incubate at room temperature for no more than 5 min or until the lysate appears clear.

> ***Critical:*** Do not vortex or pipette to avoid shearing chromosomal DNA.

(4.) Add 350 µL Buffer N3. Mix gently by inversion 6–8 times.

> ***Critical:*** Ensure complete neutralization to precipitate protein and chromosomal DNA. If Buffer P2 contains thymolphthalein, mix until the solution is colorless.

(5.) Centrifuge at $17\,000 \times g$ for 3 min to pellet precipitated material. Repeat if the lysate remains turbid.

(6.) Transfer the clear supernatant to a fresh tube for purification. Avoid any carryover of precipitate.

A › *Purification of plasmid DNA over a silica matrix*

- ☐ Buffer PB (Binding buffer), 250 mL ⟨*R*⟩
- ☐ Buffer PE (Wash buffer), 250 mL ⟨*R*⟩
- ☐ 5 mM Tris hydrochloride

(1.) Apply up to 700 µL of supernatant to a silica spin column. Centrifuge at $11\,000 \times g$ for 1 min, or use a vacuum manifold.

> ***Hint:*** Silica membranes typically bind 15–25 µg DNA. They may be reused to purify more of the same DNA.

(2.) *Optional:* Wash the silica matrix with 450 µL Buffer PB.

> ***Hint:*** Recommended if plasmids are isolated from strains containing high endonuclease activity such as HB101 or JM109.

(3.) Wash with 650 µL Buffer PE.

(4.) Dry the silica matrix by centrifugation at $11\,000 \times g$ for 2 min.

(5.) Place the spin column into a clean microcentrifuge tube.

(6.) Apply 25–55 µL 5 mM Tris pH 8.0 to the membrane. Incubate for 1 min, then elute the plasmid by centrifugation at 11 000 × *g* for 1 min.

> ***Hint:*** For higher yield or large plasmids, warm elution buffer to 70 °C.

(7.) *Optional:* Repeat elution using the same or half the volume to improve yield. Pool eluates if desired.

B › *Purification by alcohol precipitation*

- ☐ Isopropyl alcohol
- ☐ Buffer PE (Wash buffer)
- ☐ 5 mM Tris

(1.) Transfer up to 700 µL of supernatant to a clean microcentrifuge tube.

(2.) Add an equal volume of isopropanol to precipitate the DNA. Vortex to mix thoroughly.

(3.) Centrifuge at 11 000 × *g* for 5 min. Discard the supernatant and briefly spin to remove residual solvent. The pellet will appear nearly transparent.

(4.) Wash with 650 µL Buffer PE or freshly prepared 70% ethanol without disturbing the pellet.

(5.) Air-dry the pellet for 2–5 min, or dry in a vacuum centrifuge for 1 min.

(6.) Resuspend in 25–55 µL 5 mM Tris pH 8.0 or water.

C › *Purification of plasmid DNA by anion-exchange chromatography*

- ☐ Buffer P3 (Neutralization buffer), 250 mL ⟨*R*⟩
- ☐ Buffer QBT (Equilibration buffer), 250 mL ⟨*R*⟩
- ☐ Buffer QC (Wash buffer), 250 mL ⟨*R*⟩
- ☐ Buffer QF (Elution buffer), 250 mL ⟨*R*⟩

(1.) Equilibrate the anion-exchange column with the recommended volume of Buffer QBT and allow it to drain by gravity flow.

> ***Hint:*** Buffer and lysate volumes scale with tip size: roughly 4 mL per step for a 100 µg tip, 10 mL for a 500 µg tip, and 35 mL for a 2500 µg tip.

(2.) Apply the P3-neutralized, cleared lysate to the equilibrated column and let it pass through by gravity.

> ***Critical:*** The column must not run dry between steps; if flow stalls, refill with Buffer QC and continue.

(3.) Wash the column twice with the recommended volume of Buffer QC.

(4.) Elute the plasmid DNA with the recommended volume of Buffer QF into a clean tube.

(5.) Precipitate the DNA by adding 0.7 vol isopropanol at room temperature, wash, and resuspend as described in *Alternative B*.

Analyses

○ Measure DNA concentration and purity by UV absorbance at 260 nm, 280 nm, and 230 nm.

Nucleic acid	A260 = 1.0	A260/A280	A260/A230
dsDNA	50 ng/µL	1.8–1.9	2.0–2.2

> *Quality assurance:* Extinction coefficients vary with nucleotide composition. A260/A280 ratios increase with pH and vary by 0.2–0.3 between instruments. RNA contamination elevates the ratio; protein or phenol contamination depresses it.

○ Confirm plasmid identity by restriction digestion SOP 0033 and agarose gel electrophoresis SOP 0032.

> *Hint:* Use one or more enzymes predicted to linearize the plasmid or release a diagnostic fragment.

Troubleshooting

Alkaline lysis

In Step 4:

- Chromosomal DNA does not form a white precipitate but remains viscous.
  - Reduce culture volume and repeat the preparation with gentler handling.

In Step 6:

- Chromosomal DNA in the eluate
  - Use cultures within 12–16 h post-inoculation. Older cultures release degraded chromosomal DNA. If not processing immediately, store the bacterial pellet at −20 °C.
  - Mix lysate gently after adding Buffer P2. Limit lysis to 5 min to avoid DNA shearing.
  - After adding Buffer N3, mix immediately and gently to allow complete neutralization.
- RNA contamination in eluate
  - Verify RNase A was added to Buffer P1 before use. Supplement fresh RNase if needed.

Purification of plasmid DNA over a silica matrix

In Step 6:

- Little or no DNA in eluate
  - Ensure elution buffer is low salt and within pH 7.0 and pH 8.5.

Purification by alcohol precipitation

In Step 3:

- No visible pellet after isopropanol precipitation
  - Centrifuge for longer (10 min µN t). Mark the tube orientation before spinning to locate it on the expected side.
  - Add 0.1 vol of 3 M sodium acetate pH 5.2before adding isopropanol to improve precipitation of dilute samples.

Purification of plasmid DNA by anion-exchange chromatography

In Step 4:

- Low yield after elution
  - Pre-warm Buffer QF to 50 °C for large plasmids (>10 kb) to improve recovery.

Recipes

Buffer P1 (Resuspension buffer), pH 8.0

Amount	Ingredient		Stock	Final
12.5 mL	Tris-Cl, pH 8.0	R0056	1 M	50 mM
10.0 mL	EDTA, pH 8.0	R0017	0.5 M	10 mM
250 µL	RNase A, bovine pancreas	R0040	7 U/µL	7 U/L
To 250 mL	Water, reagent-grade			

Store at 4 °C. ***Critical:*** If the buffer is older than six months, supplement with fresh RNase A!

Buffer P1 (Resuspension buffer)

50 mM Tris-Cl, 10 mM EDTA, 7 U/L RNase A, pH 8.0

Date: Sign: R0006

Buffer P2 (Lysis buffer)

Amount	Ingredient		Stock	Final
2.0 g	NaOH	[1310-73-2]	40.00 g/mol	200 mM
12.5 mL	SDS	R0047	20%	1%
125 mg	Thymolphthalein □	[125-20-2]	430.54 g/mol	0.05%
To 250 mL	Water, reagent-grade			

This is why: Thymolphthalein serves as pH indicator to monitor the neutralization reaction with N3; it is optional.

Buffer P2 (Lysis buffer)

200 mM NaOH, 1% SDS, □ 0.05% Thymolphthalein

DANGER

Skin corrosion; Serious eye damage; Corrosive to metal

Date: Sign: R0007

Buffer N3 (Neutralization buffer), pH 4.8

Amount	Ingredient		Stock	Final
100.0 g	Guanidine · HCl	[50-01-1]	95.53 g/mol	4.2 M
22.0 g	KOAc	[127-08-2]	98.14 g/mol	0.9 M
To 250 mL	Water, reagent-grade			

Adjust pH with acetic acid. Store at room temperature. ***Hint:*** For use with silica-gel membranes as in the QIAGEN QIAprep Spin Miniprep Kit.

Buffer N3 (Neutralization buffer)

4.2 M Guanidine, 0.9 M KOAc, pH 4.8

DANGER

Skin corrosion; Serious eye damage

Date: Sign: R0009

Buffer PB (Binding buffer), pH 4.8

Amount	Ingredient		Stock	Final
119.0 g	Guanidine · HCl	[50-01-1]	95.53 g/mol	5.0 M
75 mL	Isopropyl alcohol	[67-63-0]	100%	30%
To 250 mL	Water, reagent-grade			

Adjust pH with acetic acid before adding isopropyl alcohol. Store at room temperature.

Buffer PB (Binding buffer)

5.0 M Guanidine, 30% Isopropyl alcohol, pH 4.8

DANGER

Flammable liquid; Central nervous system toxicity upon single exposure; Skin irritation; Eye irritation

Date: Sign: R0010

Buffer PE (Wash buffer), pH 7.4

Amount	Ingredient		Stock	Final
2.5 mL	Tris-Cl, pH 7.4	R0055	1 M	10 mM
200 mL	Ethanol, reagent-grade	[64-17-5]	100%	80%
To 250 mL	Water, reagent-grade			

Store at room temperature.

Buffer PE (Wash buffer)

10 mM Tris-Cl, 80% Ethanol, pH 7.4

DANGER

Central nervous system toxicity upon single exposure; Flammable liquid; Heart, liver, kidney, blood toxicity upon repeated exposure

Date: Sign: R0011

Buffer P3 (Neutralization buffer), pH 5.5

Amount	Ingredient		Stock	Final
73.0 g	KOAc	[127-08-2]	98.14 g/mol	2.9 M
To 250 mL	Water, reagent-grade			

Adjust pH with acetic acid. Store at 4 °C. ***Hint:*** For use with anion-exchange columns as in the QIAGEN Plasmid Kit.

Buffer P3 (Neutralization buffer)

pH 5.5

DANGER

Skin corrosion; Serious eye damage

Date: Sign: R0008

Buffer QBT (Equilibration buffer), pH 7.0

Amount	Ingredient		Stock	Final
10.95 g	NaCl	[7647-14-5]	58.44 g/mol	750 mM
2.62 g	MOPS	[1132-61-2]	209.26 g/mol	50 mM
37.5 mL	Isopropyl alcohol	[67-63-0]	100%	15%
375 µL	Triton™ X-100	[9002-93-1]	100%	0.15%
To 250 mL	Water, reagent-grade			

Adjust pH with NaOH before adding isopropanol. Store at room temperature.

Buffer QBT (Equilibration buffer)

750 mM NaCl, 50 mM MOPS, 15% Isopropyl alcohol, 0.15% Triton™ X-100, pH 7.0

DANGER

Flammable liquid; Central nervous system toxicity upon single exposure; Skin irritation; Eye irritation

Date: Sign: R0197

Buffer QC (Wash buffer), pH 7.0

Amount	Ingredient		Stock	Final
14.61 g	NaCl	[7647-14-5]	58.44 g/mol	1.0 M
2.62 g	MOPS	[1132-61-2]	209.26 g/mol	50 mM
37.5 mL	Isopropyl alcohol	[67-63-0]	100%	15%
To 250 mL	Water, reagent-grade			

Adjust pH with NaOH before adding isopropanol. Store at room temperature.

Buffer QC (Wash buffer)

1.0 M NaCl, 50 mM MOPS, 15% Isopropyl alcohol, pH 7.0

DANGER

Flammable liquid; Central nervous system toxicity upon single exposure; Skin irritation; Eye irritation

Date: Sign: R0198

Buffer QF (Elution buffer), pH 8.5

Amount	Ingredient		Stock	Final
18.26 g	NaCl	[7647-14-5]	58.44 g/mol	1.25 M
12.5 mL	Tris-Cl, pH 8.5	R0055	1 M	50 mM
37.5 mL	Isopropyl alcohol	[67-63-0]	100%	15%
To 250 mL	Water, reagent-grade			

Store at room temperature.

Buffer QF (Elution buffer)

1.25 M NaCl, 50 mM Tris-Cl, 15% Isopropyl alcohol, pH 8.5

DANGER

Flammable liquid; Central nervous system toxicity upon single exposure; Skin irritation; Eye irritation

Date: Sign: R0199

List of references

H. Birnboim and J. Doly, *Nucleic Acids Res.* **7**(6), 1513—1523 (1979).

Change log

2016-03-15	Rahul Patharkar	Alkaline lysis with ethanol precipitation
2023-05-30	Benjamin C. Buchmuller	Adaptation as SOP